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Studierichting: Informatica
Oefeningenreeks 5D nr. 1.6

$$f(x) = \frac{x}{x^2 - 1} \text{ op } \left[-\frac{1}{2}, \frac{1}{3}\right]$$

$$S = \int_a^b |f(x)| dx$$

$$S = \int_{-\frac{1}{2}}^0 \frac{x}{x^2 - 1} dx - \int_0^{\frac{1}{3}} \frac{x}{x^2 - 1} dx$$

$$\text{Stel } u = x^2 - 1$$

$$du = 2x dx \Rightarrow x dx = \frac{1}{2} du$$

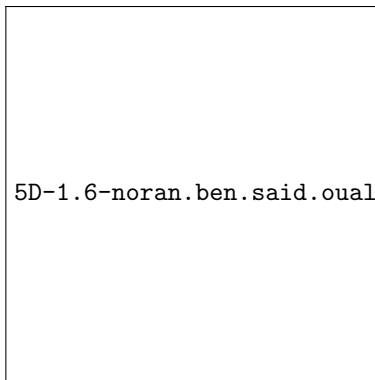
$$\int \frac{x}{x^2 - 1} dx = \frac{1}{2} \int \frac{1}{u} du$$

$$= \frac{1}{2} \ln |u|$$

$$= \frac{1}{2} \ln |x^2 - 1|$$

Dus:

$$S = \left[\frac{1}{2} \ln |x^2 - 1| \right]_{-\frac{1}{2}}^0 - \left[\frac{1}{2} \ln |x^2 - 1| \right]_0^{\frac{1}{3}}$$
$$= \frac{1}{2} \ln \frac{3}{2}$$



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References